

Data Structures

Laboratory Sheet 6

Engineering - Instituto Superior de Engenharia de Coimbra

1 - Consider that you want to implement software that offers a printer management service. Each printer is identified by a printer class that contains the relevant information:

- Name
- Port
- Brand
- Model

create a class `PrinterManager`, using the `Map` from the `java collections` library to support the operation of the following methods

- a) Printer information should be retrievable through a method with prototype `getPrinter(name)`.
- b) A method `addPrinter(Printer)` adds a new printer to the system.
- c) `getPrinterNames ()` should return a set with the names of all printers.
- d) `removePrinter (name)` should remove the identified printer
- e) boolean `existsPrinter (Name)` indicates whether the specified printer is registered or not.

2 Consider that you can send jobs to each of the printers (described in the previous paragraph), through a `sendJob` method. These jobs are characterized by the following information

- file name
- home page
- final page.

The printer also offers class `hasNextJob ()` and `nextJob()`, which are used by another component of the system (which you do not have to implement and is not your concern) for getting from each `Printer` object the next job to be printed.

It is intended jobs are queued on each printer, in order of size: that is, a job with fewer pages will always have priority over print a job with more pages.

Using appropriately the class *PriorityQueue*,

- a) Implement the classes and methods described, as well as any others that are necessary.
- b) Implement a method `print(Job)` in `PrinterManager` class that allocates a job to a printer according to the following criteria:
 - a. The printer with less jobs
 - b. In case of a tie, the printer is chosen whose next job to be printed has fewer pages.
 - c. Should the tie persist, the tie can be broken arbitrarily.